

Unless specified otherwise, assume the working fluid is air and it can be treated as an ideal gas with constant specific heats $\gamma=1.4$. The specific gas constant for air is:

$$R = 287 \text{ J/(kg} \cdot \text{K)} \text{ in SI units or } R = 1716 \text{ ft} \cdot \text{lbf / (slug} \cdot \text{R)} \text{ in English units.}$$

In solving the problems below, unless otherwise specified, you may use the online compressible calculator <https://devenport.aoe.vt.edu/aoe3114/calc.html>

1. Consider a nozzle with $A_e/A_{th} = 4$, reservoir pressure $p_{res} = 20 \text{ atm}$, and temperature $T_{res} = 600 \text{ K}$. Calculate the Mach number, pressure, and temperature at the throat and the exit for the following cases:

- (a) The flow is supersonic at the exit (perfectly expanded flow).
- (b) The flow is subsonic for the entire nozzle except at the throat, where it is sonic (choked subsonic flow $M_{throat} = 1$)
- (c) Calculate the required reservoir pressure for choked subsonic flow when the exit pressure is $p_{exit} = 0.468 \text{ atm}$.

2. A nozzle is designed to expand air from a chamber in which the pressure and temperature are 2000 KPa and 60°C respectively to a Mach number of 4.75. The throat area of the nozzle is 0.025 m^2 . Find

- a. The exit area of the nozzle
- b. The mass flow rate through the nozzle when operating at design conditions (supersonic exit)
- c. The back pressure for which it will be a normal shock wave on the exit plane of the nozzle
- d. The range of back pressures where there will be a normal shock wave in the nozzle
- e. The range of back pressures over which there will be oblique shocks at the exit (overexpanded nozzle)
- f. The range of pressures over which there will be expansion waves outside the nozzle (under-expanded nozzle)

3- Air flow through a frictionless adiabatic C-D nozzle. The air stagnation pressure and temperature are $7.0 \times 10^5 \text{ N/m}^2$ and 500 K, respectively. The diverging portion of the nozzle has an area ratio of $A_{exit}/A_{throat} = 11.91$. The exit pressure is around $2.3 \times 10^5 \text{ N/m}^2$. Find the location of the shock, determine the Mach number, the static pressure and the temperature at the exit plane of the nozzle.